



DIUMED

DiuMed

Clinical-Grade Health Screening — No Extra Hardware, No Internet Required

VMEDITHON 3.0

OPEN INNOVATION TRACK

INSTITUTION

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TEAM

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PROBLEM STATEMENT

The Healthcare Access Gap in Rural India

Millions of rural mothers, low-income households, and frontline ASHA workers cannot get basic clinical screening for anemia, jaundice, and cardiac risk — not because a cure or test does not exist, but because cost, connectivity, and privacy stand in the way of reaching them.



Hardware Barrier

ROOT CAUSE

Clinical screening today requires ₹5,000+ smartwatches, pulse oximeters, or diagnostic devices.

CONSEQUENCE

Rural mothers & low-income households are excluded entirely — the tool itself is the barrier.



Network Blackouts

ROOT CAUSE

Existing health apps are cloud-dependent, sending readings to a server for processing.

CONSEQUENCE

The moment 4G/5G drops in remote regions, the app — and the screening — fails completely.



Privacy & Surveillance Risk

ROOT CAUSE

Centralized platforms stream sensitive patient telemetry to external servers, often unencrypted.

CONSEQUENCE

Patients lose trust and control over their own health data, discouraging adoption at the source.

WHO IS LEFT BEHIND



Rural mothers & low-income households



Primary health centers with no reliable internet



ASHA workers in remote field settings

SOLUTION

One Smartphone. Zero Extra Hardware. Fully Offline.

DiuMed turns any standard smartphone into an offline-first, AI-powered health OS — using only the camera and flash.



rPPG Heart Rate & SpO₂

Camera-based remote photoplethysmography detects vitals — no contact sensor needed.



Ocular AI Screening

Detects anemia and jaundice from eye-region analysis in real time.



Offline SMS Alerts

Cryptographically signed emergency alerts sent via SMS — no internet required.



AES-GCM 256 Vault

All patient data encrypted locally. Zero-trust architecture by design.

HOW IT WORKS — ON-DEVICE PIPELINE



Capture

Camera + flash reads skin & eye signal



Preprocess

Signal cleaned & normalized on-device



AI Inference

On-device ML scores vitals & risk



Alert & Vault

Signed SMS alert + encrypted storage

SOCIAL IMPACT

Bringing Health Awareness to Rural India



Empowering ASHA workers as the bridge between DiuMed and the village — trusted local hands deliver a new screening tool.

No literacy or tech-fluency barrier: a phone camera and a one-minute scan replace complex devices and app logins — making health checks approachable for first-time users.

01

Earlier Detection

Anemia, jaundice & cardiac risk caught before escalation — awareness camps flag risk early instead of after symptoms worsen.

02

Zero Hardware Cost

Screening on any existing Android smartphone removes the ₹5,000+ affordability barrier for rural families.

03

Life-Saving Alerts

Offline SMS keeps care flowing even without internet — reaching the last mile where awareness matters most.

SCALABILITY & FEASIBILITY

Built to Ship on Real Hardware

TECHNOLOGY STACK

Layer	Tool / Library
ML Framework	TensorFlow Lite / ML Kit
Camera API	Standard Android Camera API
Messaging	SMS Manager API
Encryption	PBKDF2 + AES-GCM
Deployment	Low-cost Android devices

CURRENT PROTOTYPE PROGRESS

-  rPPG camera capture module operational
-  Ocular region detection pipeline implemented
-  Basic on-device ML inference running end-to-end



Scales Without Scaling Cost

Runs on any low-cost Android phone — no proprietary hardware, so every new device is a new screening point at zero marginal cost.

NEXT 6–12 MONTHS



Clinical Validation

Field testing with ASHA workers in live rural settings.



Accuracy Testing

Model validation across skin tones & lighting conditions.



Pilot Deployment

Rollout to rural Primary Health Centers (PHCs).



Regulatory Compliance

Pursue certifications & healthcare regulatory approvals.



Thank You

One camera. Zero barriers. Lifesaving care for every corner of India.

TEAM DIUFOUNDERS

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